Civics Group	Index Number	Name (use BLOCK LETTERS)

ST. ANDREW'S JUNIOR COLLEGE
2011 JC2 Preliminary Examinations

H1 BIOLOGY **8875/1**

Paper 1: Multiple Choice

Wednesday 21 September 2011 1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet only.

This document consists of **13** printed pages.

1 The correct shape of a globular protein is maintained by several types of bond.

Bond types	1	disulphide
	2	hydrogen
	3	ionic
	4	peptide

Which bond types are involved in maintaining the different levels of structural complexity of a globular protein?

	Level of structural complexity		
	Primary	Secondary	Tertiary
A	4	3	1, 2, 3
B	2	1	2, 3, 4
C	2	4	2, 3, 4
D	4	2	1, 2, 3

2 Three pure substances are analysed. The table shows the elements that each contains.

	C	H	O	N	P	S
X	+	+	+	+	-	-
Y	+	+	+	-	-	-
Z	+	+	+	+	-	+

Which combination could be correct?

	X	**Y**	**Z**
A	adenine	carbohydrate	fat
B	adenine	fat	an amino acid
C	an amino acid	fat	carbohydrate
D	fat	carbohydrate	an amino acid

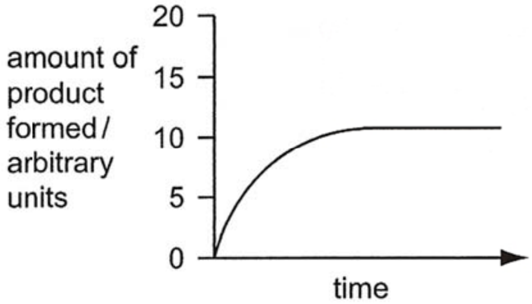
3 The list shows three characteristics of enzyme activity.

1 Reaction rate decreases if the concentration of non-competitive inhibitor increases.
2 Reaction rate is reduced at extremes of pH.
3 Reaction rate is reduced at low temperature.

What explains each of these characteristics?

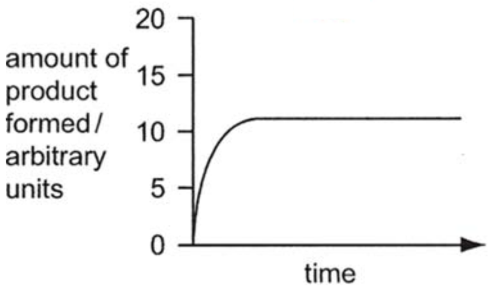
	Availability of active site is reduced.	Reduced kinetic energy reduces the rate of molecular collisions.	Hydrogen bonding is disrupted.
A	1	2	3
B	2	1	3
C	1	3	2
D	2	3	1

4 The graph shows the amount of product formed by a standard concentration of enzyme and a standard concentration of substrate at a temperature of 15°C.

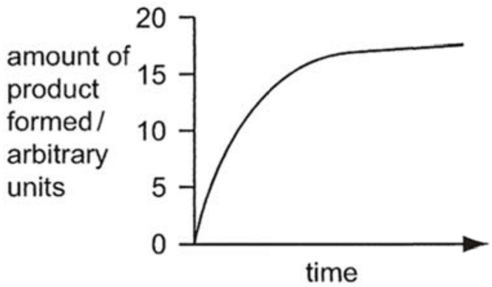


Which graph shows the effect on the activity of the enzyme of increasing the temperature to 20°C?

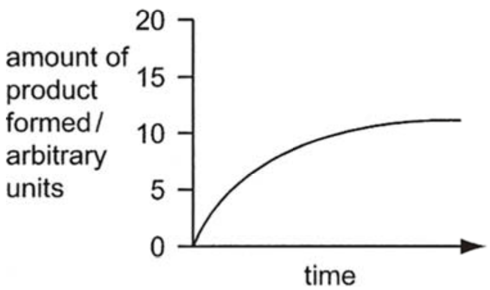
A



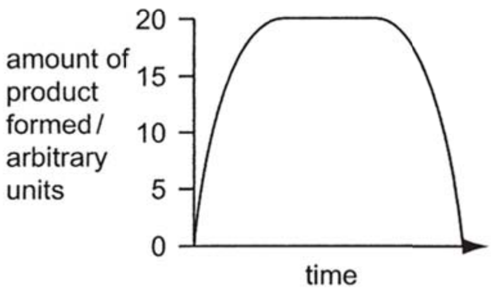
B



C



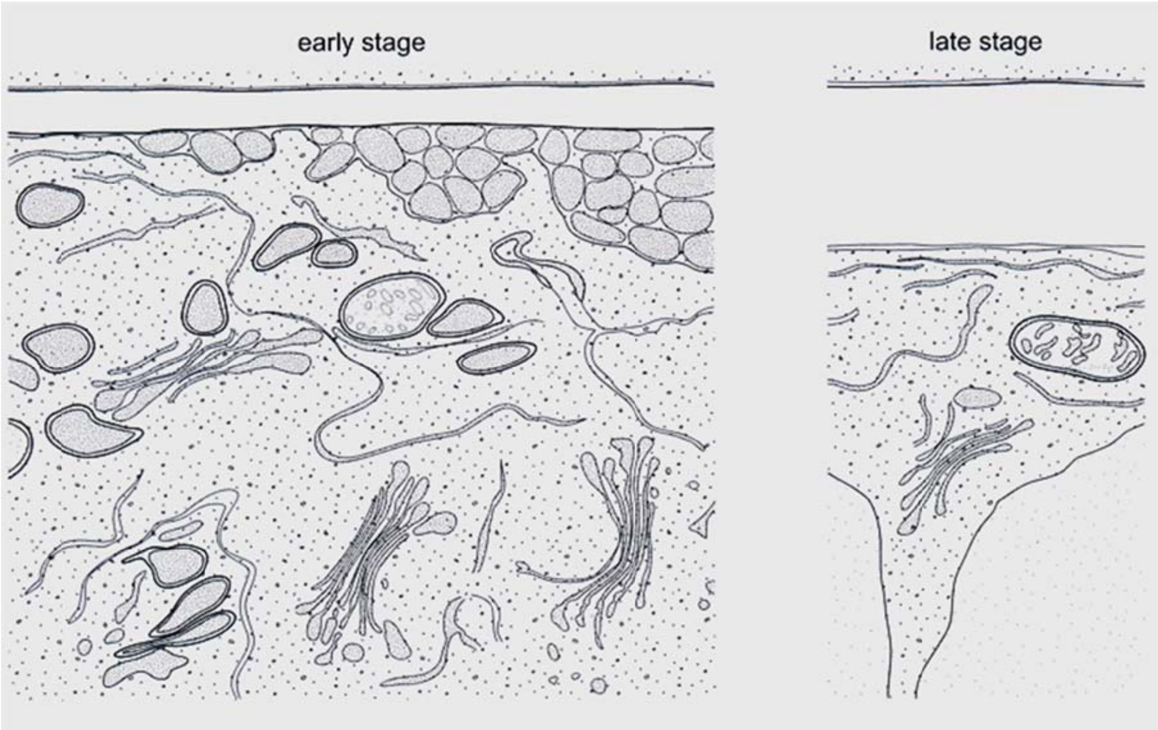
D



5 A severe inherited condition arises from the failure to produce an enzyme that breaks down glycoproteins in cells. The condition can be diagnosed from an electron micrograph of a patient's cells. Which abnormality would be observed in these cells?

A An incomplete chromosome due to the lack of a gene.
B Larger lysosomes due to the accumulation of glycoproteins.
C Thinner cell surface membrane due to the lack of glycoproteins.
D Less endoplasmic reticulum due to a reduction in protein synthesis.

6 The photo electron micrographs show early and late stages in the development of the cell wall in a young plant cell.



Which statement describes the events leading to the development of the cell wall?

A Complex carbohydrates assembled in the Golgi body are exported to the cell wall by the Golgi vesicles.
B Enzymes in the cell surface membrane synthesise the cell wall components from soluble carbohydrates brought by the Golgi vesicles.
C Polysaccharides are exported to the cell wall and synthesized into wall components by the Golgi body.
D Ribosomes synthesise glycoproteins that are exported by Golgi vesicles to be used in the cell wall.

7 A certain cell surface membrane is made up entirely of phospholipids and is 4nm thick. A volume of 1 mm^3 of this membrane was homogenized and dropped onto a large tray of water. The phospholipids spread out to form a continuous thin film.

What is the expected area of this film?

A 250 000 mm^2 because the molecules originally exist as a single layer
B 250 000 mm^2 because the molecules originally exist as a double layer
C 500 000 mm^2 because the molecules originally exist as a single layer
D 500 000 mm^2 because the molecules originally exist as a double layer

8 After a liver cell containing 12 chromosomes undergoes mitosis, one daughter cell contains 11 chromosomes and the other contains 13 chromosomes.
At which stage in mitosis has an error occurred?

A Prophase
B Metaphase
C Anaphase
D Telophase

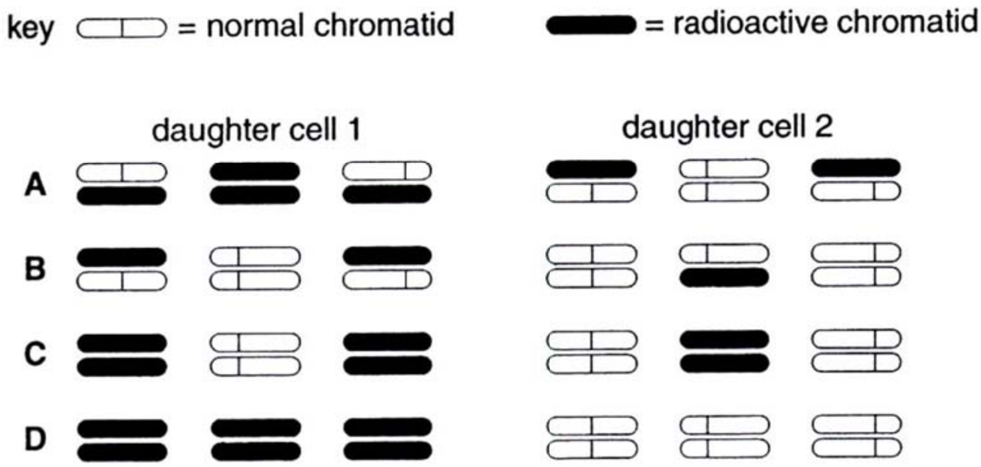
9 The amount of DNA in a skin cell of a mouse at the end of interphase is 12pg.
What will be the amount of DNA in a sperm cell at metaphase **II**?

A 3pg
B 6pg
C 9pg
D 12pg

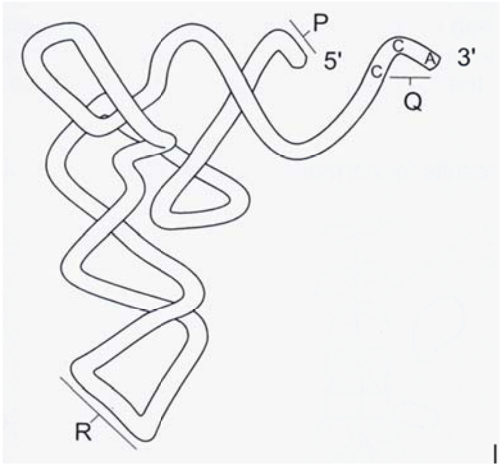
10 The following experiment was carried out.

1. Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive thymine, so that all the DNA was labelled.
2. Cells in early interphase were then transferred to a medium where the available thymine was not radioactive.
3. A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?



11 The diagram shows a tRNA molecule.



What are the roles of the parts of the molecule labelled P, Q and R?

	Binding of mRNA	Binding of amino acids	Binding of peptide
A	R	Q	P
B	P	R	Q
C	Q	P	R
D	P	Q	R

12 In planet Y, there is an alternative genetic code. Assuming that there is a total of 81 amino acids coded for, and each codon consists of 4 bases, how many different types of bases are available?

A 3
B 4
C 5
D 6

13 Which of the following mutations would lead to the occurrence of cancer?

1 germline mutation in a copy of p53 gene
2 somatic mutation in second copy of p53 gene
3 gene amplification of Ras gene
4 point mutation in a copy of Ras gene

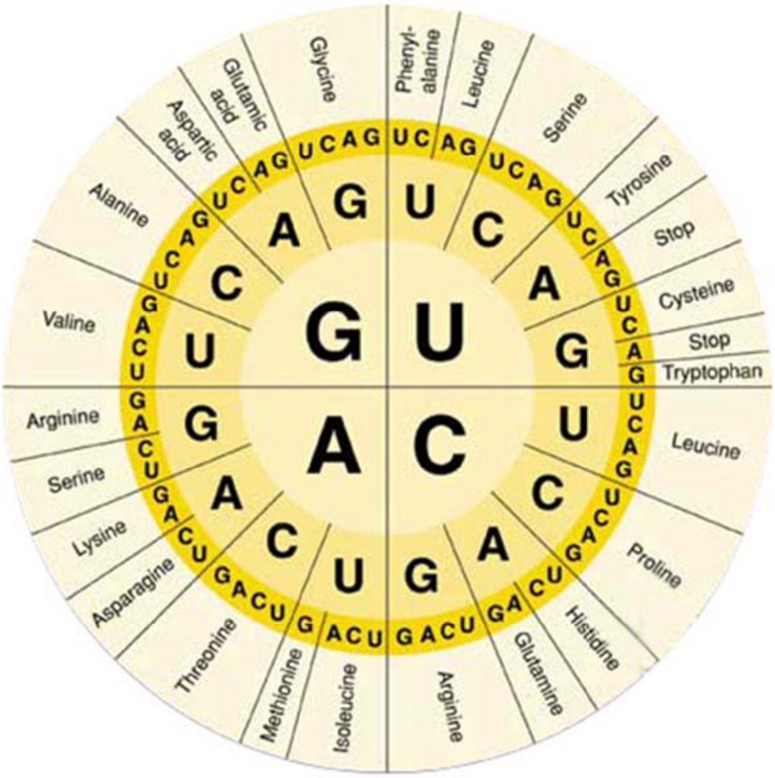
A 1 and 4
B 2 and 3
C 2, 3 and 4
D 1, 2, 3 and 4

14 A pentapeptide has the following amino acid sequence :

H_2N – Met – Leu – Trp – Ala – Phe – COOH

Where

Met : Methionine
Leu : Leucine
Trp : Tryptophan
Ala : Alanine
Phe : Phenylalanine



Using the genetic code shown in the diagram above, deduce the corresponding template strand sequence during transcription.

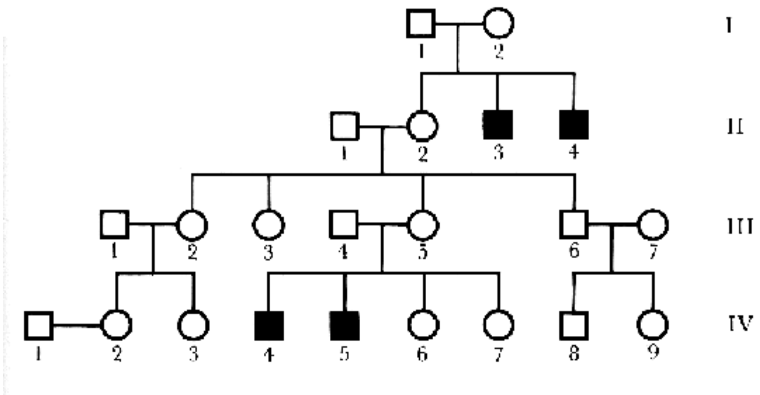
A 5' – GAA GGC CCA CAA CAT – 3'
B 5' – TTA GAA GGC CCA CCA CAT – 3'
C 5' – ATG TTG TGG GCC TTT TAG – 3'
D 5' – AUG CUG UGG GCA UUC – 3'

15 Down’s syndrome can be caused by two forms of chromosome aberration. Classic Down's syndrome is a trisomy of chromosome 21 that affects all cells. A mosaic condition occurs when there are two or more cell types in the body with differences in chromosome number and structure.

Suggest how the mosaic condition arises?

A non-disjunction of chromosomes in mitosis in early fetal development
B non-disjunction of chromosomes in meiosis during formation of the ovum
C non-disjunction of chromosomes in meiosis during formation of sperm
D translocation of chromosomes at maturation of the ovum

16 Haemophilia is a rare genetic disorder in which the blood does not clot normally. The mutation that causes haemophilia is located on the X-chromosome. The pedigree below shows the inheritance of haemophilia in a family. Squares represent males while circles represent females. Dark symbols represent affected individuals.



What is the probability that a son born to IV-2 will be a haemophiliac?

A 0.5
B 0.25
C 0.125
D 0.0625

17 A brown-shelled snail mates with a black-shelled snail of the same species, produced F1 offspring with shells consisting of brown and black patches. Sibling mating of the F1 offspring produced an F2 generation in which 99 snails had shells with brown and black patches, 51 had brown shells and 49 had black shells.
What does this experiment demonstrate?

A codominance
B dihybrid cross
C multiple alleles
D sex linkage

18 Fruit flies *Drosophila* homozygous for long wings, were crossed with flies homozygous for vestigial wings. The F_1 and F_2 generations were raised at three different temperatures.

At each temperature, the F_1 generation all had long wings.

The table below shows the results in the F_2 generation.

Temperature	***Result***
21°C	¾ long wings, ¼ vestigial wings
26°C	¾ long wings, ¼ intermediate wing length
31°C	all long wings

Which statement explains these results?

A Wing length is under polygenic control.
B Long wing and vestigial wing illustrate codominance at 26°C.
C Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.
D Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.

19 In experiment 1, male cats with stripes are mated with female cats without stripes. In experiment 2, female cats with stripes are mated with male cats without stripes. Only true breds were used in both experiments, and stripes is a dominant trait. Which of the following statements is correct?

A The gene coding for stripes is found on an autosomal chromosome, because F_1 offspring in both experiments consists of all striped cats.

B The gene coding for stripes is found on an autosomal chromosome, because F_1 offspring in both experiments consists of striped females and males without stripes.

C The gene coding for stripes is found on the X chromosome, because F_1 offspring in experiment 1 consists of striped females and males without stripes, while F_1 offspring in experiment 2 consists of all striped cats.

D The gene coding for stripes is found on the Y chromosome, because F_1 offspring in experiment 1 consists of striped females and males without stripes, while F_1 offspring in experiment 2 consists of all striped cats.

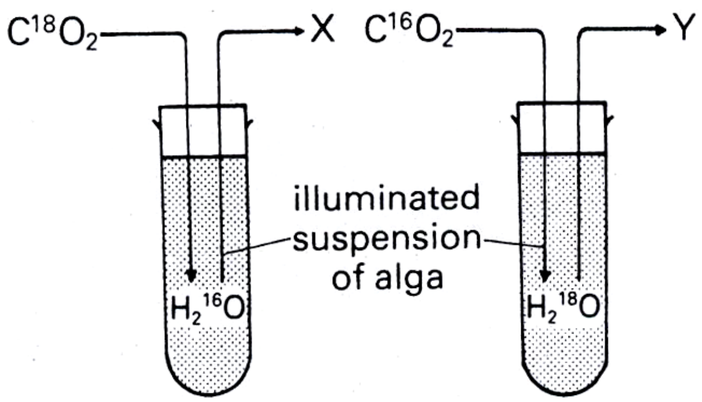
20 The table shows the amount of carbon dioxide taken up by a plant in the light and the amount of carbon dioxide produced in the dark, at different temperatures.

Temperature / °C	Net uptake of CO_2 in light / mg $g^{-1}h^{-1}$	Release of CO_2 in dark / mg $g^{-1}h^{-1}$)
5	1.1	0.2
15	2.8	0.8
30	2.0	2.4

Assuming that the rate of respiration of this plant is the same in light and dark, which of the following statement is correct.

A The true uptake of carbon dioxide in the light at 15°C is 2.0 mg $g^{-1}h^{-1}$.
B Temperature has no effect on the rate of carbon dioxide uptake in the light.
C The rate of photosynthesis exceeds the rate of respiration at all temperatures.
D None of the above

21 In the experiment shown in the diagram, the oxygen given off at **X** and **Y** will be labelled with the isotopes _____ and _____ respectively.



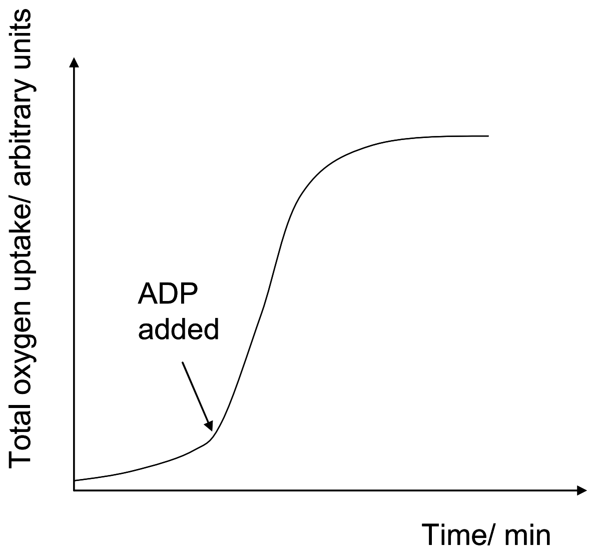
	X	Y
A	^{16}O	^{16}O
B	^{16}O	^{18}O
C	^{18}O	^{16}O
D	^{18}O	^{18}O

22 2,4-dinitrophenol is a chemical that targets the inner mitochondrial membrane to uncouple oxidative phosphorylation from electron transport. It allows protons to cross the inner mitochondrial membrane and thus dissipates the proton gradient.

Which of the following is true of a cell treated with 2,4-dinitrophenol?

	Ability to use glucose	**Ability to use oxygen**	**Ability to produce carbon dioxide**
A	Yes	Yes	Yes
B	Yes	No	Yes
C	Yes	No	No
D	No	No	No

23 The graph shows the total oxygen uptake by a sample of homogenised liver tissue over a period of time. Which statement explains the sharp rise in oxygen uptake after the addition of ADP?



A ADP reduces the molecular oxygen to water.
B Electron transport results in the oxidation of ADP.
C Electron transport requires the hydrolysis of ADP.
D Electron transport is accompanied by simultaneous phosphorylation of ADP.

24 Which of the following descriptions refers accurately to homologous structures?

	Basic structure	**Common ancestor**	**Function performed**
A	Same	No	Different
B	Different	No	Same
C	Same	Yes	Different
D	Different	Yes	Same

25 Which sequence of events correctly describes evolution?

1 Differential reproduction of the spiders occurs.
2 A new selection pressure occurs.
3 Allele frequencies within the spider population change.
4 Poorly adapted spiders have decreased survivorship.

A 2, 4, 1, 3
B 2, 4, 3, 1
C 4, 1, 3, 2
D 4, 3, 1, 2

26 A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25 % of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1:1 to 4:1.

If mice produce an average of eight offspring per litter and the pattern of predation remains the same, what would happen to the population of light mice?

A They disappear after one further generation.
B They disappear after two further generations.
C They remain at the level of one fifth of the population.
D They will be reduced to a constant but very low frequency.

27 Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

A Comparative embryology
B Homologous structures
C Studying of their ecological niches
D Molecular comparisons of DNA and amino acid sequences

28 A PCR reaction consists of three major steps: denaturing, primer annealing and elongation. A PCR reaction was performed using genomic DNA as a template without success.

Analysis of the tube contents immediately after the elongation step at the 20th cycle of the PCR reaction revealed a lack of increase in DNA amount since the start of the reaction, DNA only exist as single-stranded form with no primers attached, primers, enzymes and deoxyribonucleotides were free floating in solution.

Which of the following could be a probable explanation for these results?

A genomic DNA cannot be used as templates for PCR
B temperature for all three steps of PCR was set at 95^0C
C the enzyme used is not heat-resistant
D the primers used fail to bind to DNA fragment of interest

29 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell. Which statement explains this?

A Genes not required for a particular cell line are prevented from being expressed.
B mRNA that is not required for a particular cell line is destroyed.
C Genes not required for a particular cell line are removed using restriction enzymes.
D Different stem cells have only the genes required for their particular cell line.

30 To date, genetically modified food crops have been modified to ___________.

A delay ripening of fruits
B produce needed human nutrients
C resist herbicides
D all of the above